FastSpeech 2 for English and Telugu: Technical Report

Code, checkpoints and results: https://github.com/rashiii02/fastspeech2-english-telugu

1. Implementation

We trained two separate FastSpeech 2 models, one for English and one for Telugu, using the PyTorch reference implementation by ming024 as the base. The model is a feed-forward Transformer with a phoneme encoder, a variance adaptor (duration, pitch and energy predictors with a length regulator) and a mel-spectrogram decoder. Pitch and energy are modelled at phoneme level and quantised into 256 bins with learned embeddings. Mel-spectrograms are converted to waveforms with the pretrained universal HiFi-GAN vocoder (22.05 kHz).

Training used 20,000 steps, batch size 16, Adam (betas 0.9/0.98) with 4,000 warm-up steps, one T4 GPU per language, and losses logged every 100 steps with validation every 1,000 steps. The saved checkpoints are fp16 weights (68 MB each). The total loss is the sum of mel, post-net mel, duration, pitch and energy losses.

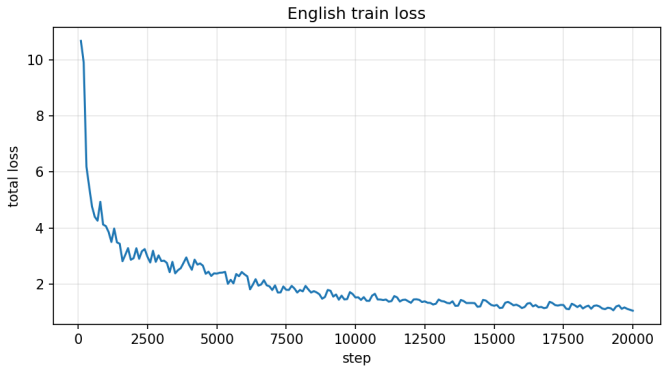
Changes to the reference code were needed only for newer libraries: updated librosa STFT arguments, the tgt interface for TextGrids (empty intervals mapped to the silence token sp), a float32 cast of the losses (a Double/Float dtype error in the backward pass), and a speaker-aware multi-speaker setup for Telugu.

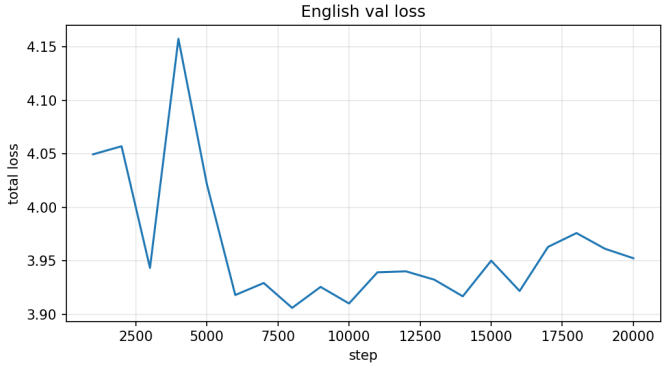
2. Data and preprocessing

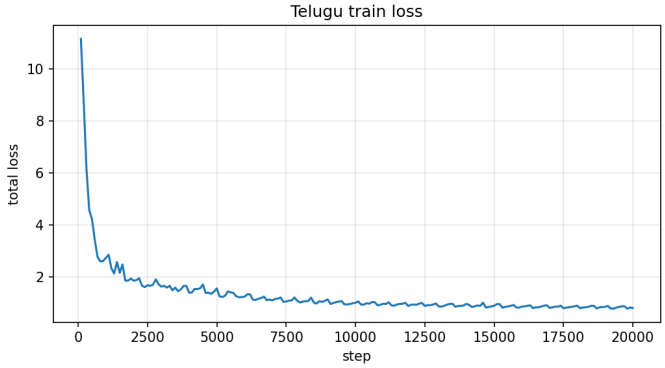
English uses the Svarah dataset (accented Indian English). The Hugging Face copy has no speaker column, so all utterances were treated as one speaker (spk0). **Telugu** uses IndicTTS Telugu with its two speakers (female and male), so the Telugu model is multi-speaker. For each language we shuffled with seed 42, kept clips between 1 and 15 s, resampled to 22,050 Hz mono, and filled the training set to 5.0 h and then the validation set to 1.0 h, with no overlap (English 3,185 train / 637 val utterances; Telugu 2,261 train / 452 val).

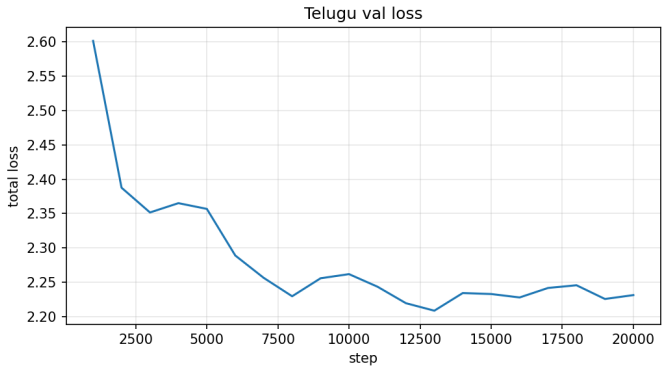
Alignments were produced with Montreal Forced Aligner 3.4.3. English used the pretrained english_us_arpa model and dictionary (3,889 of 3,924 utterances aligned). For Telugu we trained an aligner from scratch with a grapheme dictionary in which every Telugu Unicode character is one symbol (2,713 of 2,718 aligned); to fit the 12 h Kaggle limit we used 10 monophone and 10 triphone iterations instead of the default schedule. Telugu text was NFC-normalised, ZWJ/ZWNJ removed, and only the Telugu block (U+0C00 to U+0C7F) kept; each character became a symbol in the model vocabulary. Pitch (pyworld) and energy were averaged per phoneme and normalised with the dataset statistics. Mean F0 was about 176.5 Hz (std 54.6) for English and 192.8 Hz (std 43.2) for Telugu, which is consistent with the Telugu data mixing a female and a male speaker.

3. Training and validation loss









English. Training loss falls quickly from about 10.7 to about 3 in the first 2,000 steps and then declines slowly to about 1.05 at step 20,000. Validation loss stays almost flat between 3.9 and 4.16: its minimum is about 3.91 at step 8,000 and it ends at about 3.95. The gap of roughly 3 points between train and validation, and the validation curve that stops improving after about 6,000 steps, show limited generalisation. Likely reasons are that Svarah is an ASR benchmark with many different speakers and recording conditions treated as one speaker, so pitch, energy and duration targets are noisy and cannot be predicted well from text, and that the model fits the 5 h training set. More data or a real single-speaker corpus would help; we did not test this.

Telugu. Training loss drops from about 11 to about 0.8, and validation loss decreases from 2.60 to about 2.21 (minimum near step 13,000), ending at about 2.23. The gap is smaller (about 1.4) and the validation curve is smoother, which fits cleaner studio recordings from only two speakers. Validation is nearly flat after step 8,000 with small oscillations, so extra training would give little gain and may slightly overfit. Because the validation minima occur before step 20,000 in both runs, early stopping around 8,000 (English) and 13,000 (Telugu) would be slightly better; we kept the 20,000-step checkpoint for both.

4. Speech quality per test sentence

Quality notes are from listening to the generated audio in Results/*/test_generated_audios. The text front-end verbalised numbers in both languages (English: 'ten forty five A M', 'twelve November twenty twenty six'; Telugu: number words for 2026, 12 and 10:45), and inserted a silence token (sp) at the comma in E4 and T4.

English

All five English sentences are clear and intelligible with natural prosody.

ID	Test focus	Quality (listening)
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings. (long rare word)	Clear and intelligible; the very long word is pronounced correctly.
E2a	The orchestra's conduct during the ceremony was praised by the audience. (conduct, noun)	Clear; 'conduct' is stressed as a noun (first syllable).
E2b	The researchers will conduct an experiment to evaluate the proposed method. (conduct, verb)	Clear; 'conduct' is stressed as a verb (second syllable), different from E2a.
E3	The rhythm of the orchestra changed unexpectedly. (irregular spelling)	Clear; 'rhythm' is pronounced correctly.
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time. (numbers, date and an out-of-vocabulary word)	Clear; the time and date are read correctly, with a pause at the comma; 'andaroo' is read as written.

Telugu

All five Telugu sentences are clear and intelligible; pronunciation is accurate and the voice is stable.

ID	Test focus	Quality (listening)
T1	Long technical Telugu word (neuropharmacological) in a research sentence.	Clear; the long technical word is pronounced correctly.
T2a	Telugu sentence about a strong grip on a rope (pattu = grip).	Clear; 'pattu' (grip) is pronounced correctly.
T2b	Telugu sentence about wearing a silk saree to a ceremony (pattu = silk).	Clear; 'pattu' (silk) is pronounced the same way as in T2a, since the text is identical.
T3	Short Telugu sentence: the tiger walked silently in the forest.	Clear and natural.
T4	Telugu sentence with a time and date (12 November 2026, 10:45) and a comma pause.	Clear; the date and time are read correctly, with a pause at the comma.

5. Variance adaptor plots per sentence

For each sentence the plots show predicted phoneme duration (ms), phoneme-level pitch (Hz) and phoneme-level energy. Telugu phoneme labels on the x-axis are not readable because the plotting font has no Telugu glyphs; the order of phonemes is still correct.

Per-sentence observations

E1. Very long utterance (85 phonemes); the long word spans many consecutive short phonemes, and pitch is expected to drift downward over the clause.

E2a. Pitch ranges from about 118 to 247 Hz: a rise on the first content word, general downward drift (declination) across the clause, a local peak on 'praised' and the lowest values on the final word, with a small rise on the last phoneme.

E2b. Compare with E2a: the same word 'conduct' appears as a verb here, so a different stress pattern on its two syllables should appear in the duration and pitch plots.

E3. Shortest English sentence (37 phonemes); a single declination contour with little variation.

E4. Duration plot: most phonemes last 30 to 130 ms; the largest bar is the sp pause at the comma (about 420 ms), and the last phoneme before it and some stressed vowels or stops (for example a /t/ near 230 ms) are longer. This shows the duration predictor learned phrase-final lengthening and pause placement.

T1. Long technical word; many consecutive consonant-vowel units with moderate duration changes.

T2a. Compare with T2b: the same word 'pattu' occurs with different meanings, so differences in the plots come only from context.

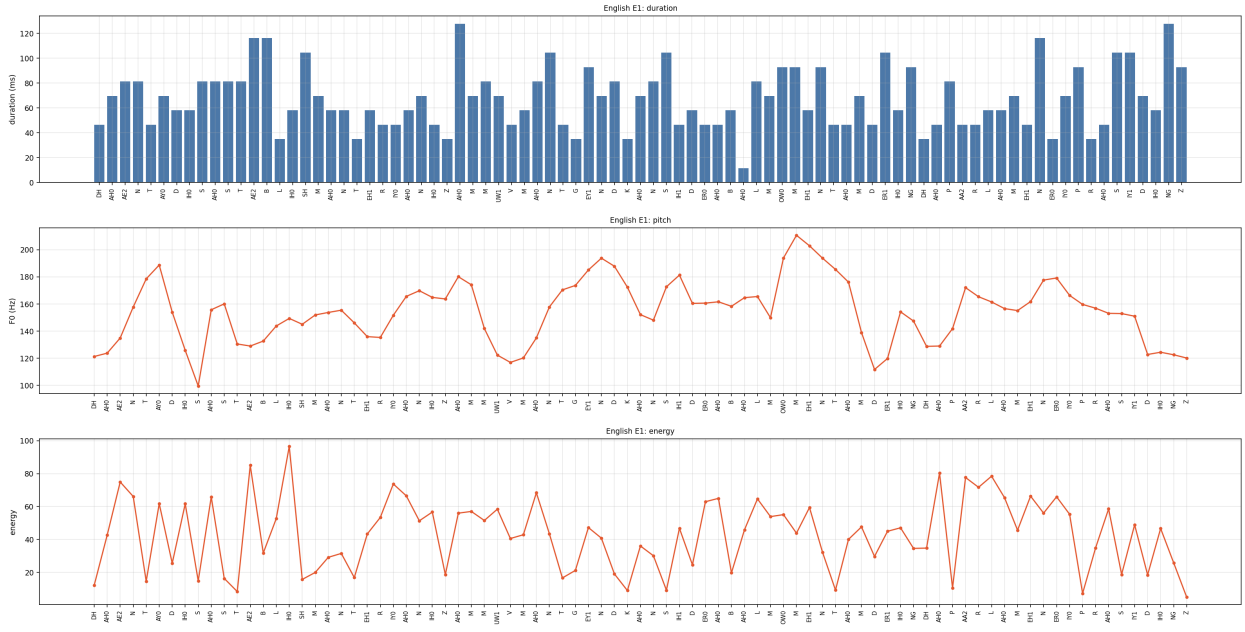
T2b. See T2a; contour should be similar for the shared word and different elsewhere.

T3. Short sentence (29 symbols) with a smooth falling pitch contour.

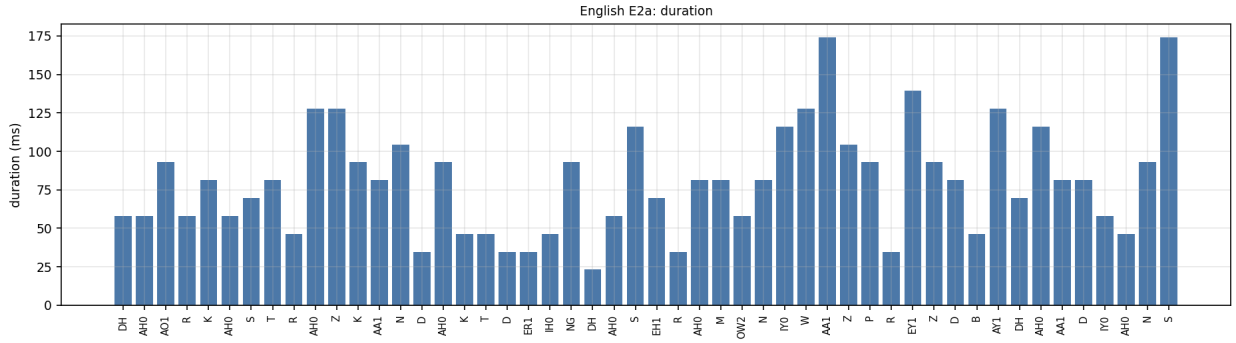
T4. Pitch ranges from about 162 to 265 Hz with many local peaks (frequent word and syllable-level variation) and a gradual downward trend; a pitch reset after the sp pause at the comma and a low final value.

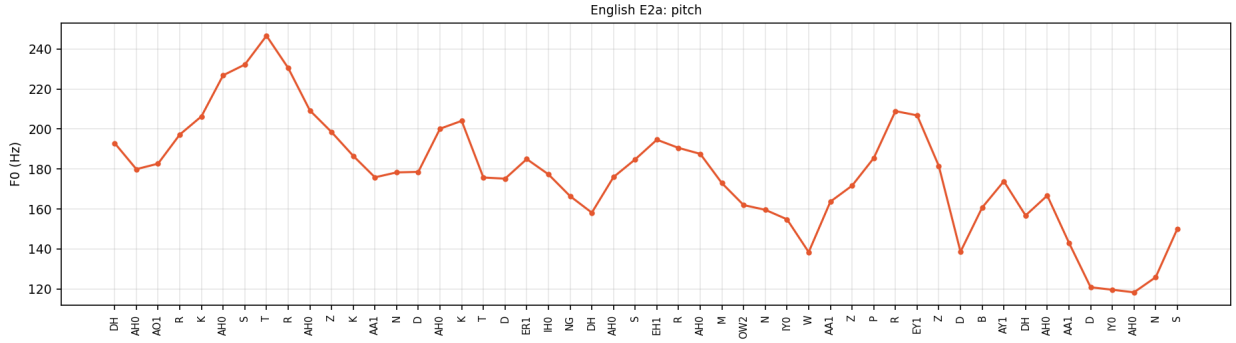
Plots

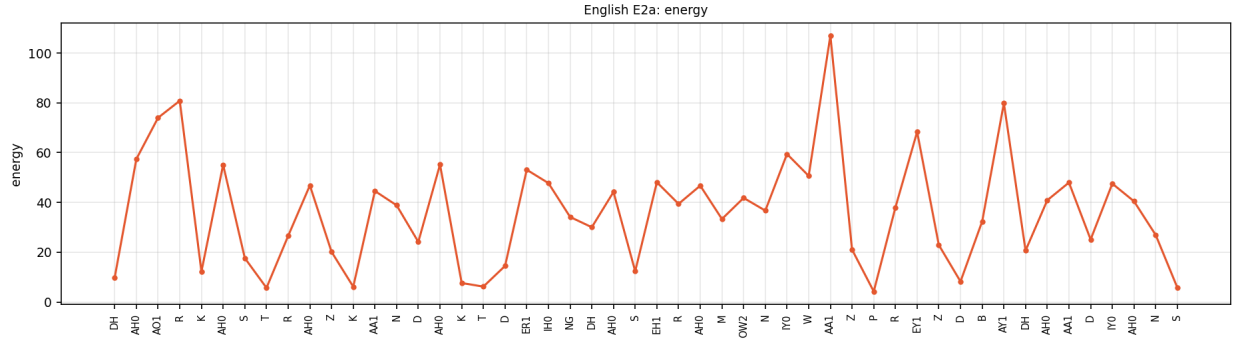
E1



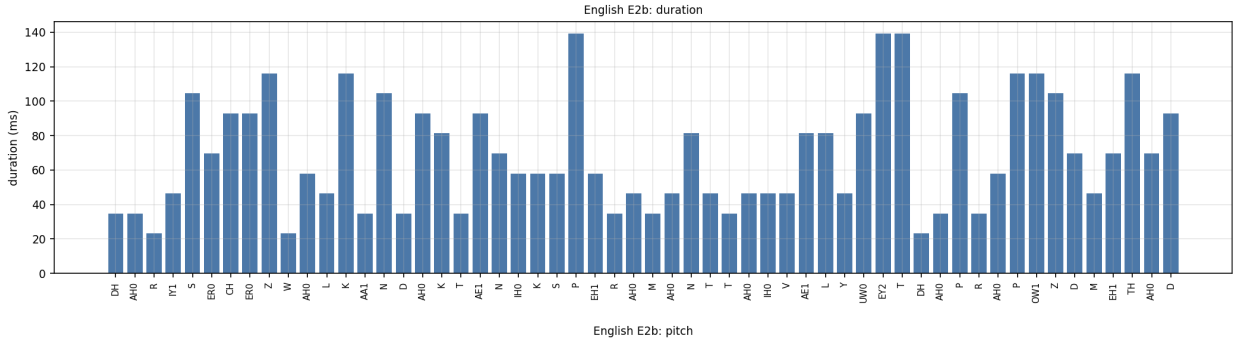
E2a

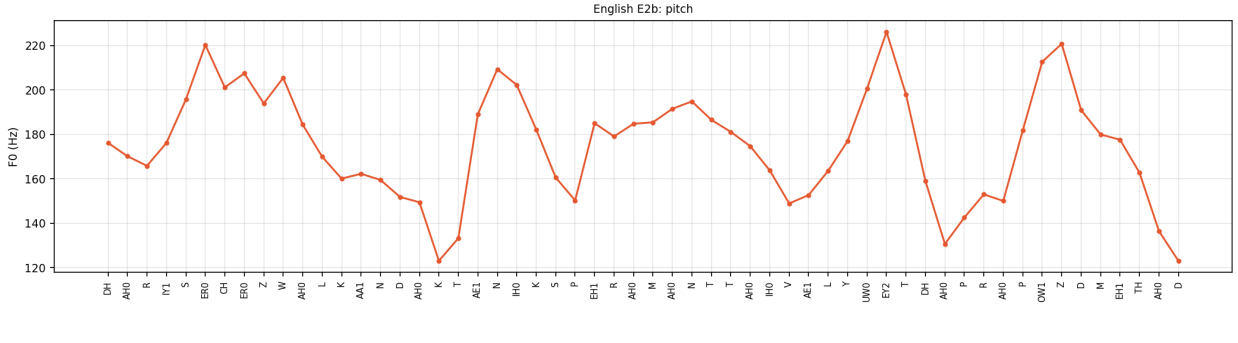

English E2a: duration
duration (ms)
0
25
50
75
100
125
150
175
DH AH0 AO1 R K AH0 S T R AH0 Z K AA1 N D AH0 K T D ER1 IH0 NG DH AH0 S EH1 R AH0 M OW2 N IY0 W AA1 Z P R EY1 Z D B AY1 DH AH0 AA1 D IY0 AH0 N S

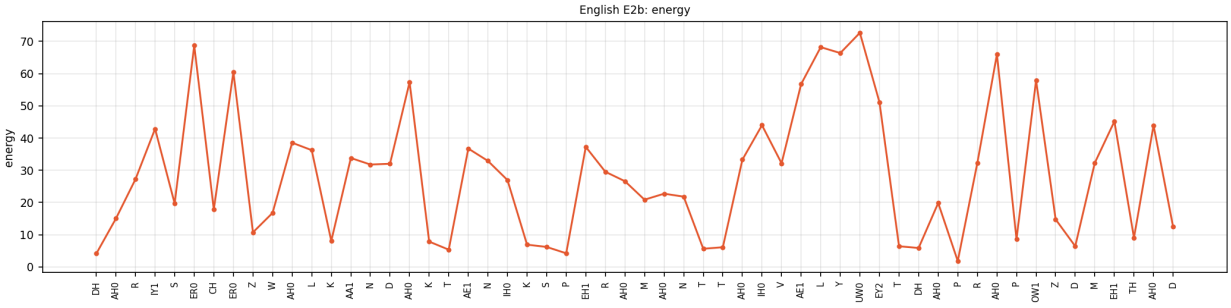

English E2a: pitch
F0 (Hz)
120
140
160
180
200
220
240
DH AH0 AO1 R K AH0 S T R AH0 Z K AA1 N D AH0 K T D ER1 IH0 NG DH AH0 S EH1 R AH0 M OW2 N IY0 W AA1 Z P R EY1 Z D B AY1 DH AH0 AA1 D IY0 AH0 N S


English E2a: energy
energy
0
20
40
60
80
100
DH AH0 AO1 R K AH0 S T R AH0 Z K AA1 N D AH0 K T D ER1 IH0 NG DH AH0 S EH1 R AH0 M OW2 N IY0 W AA1 Z P R EY1 Z D B AY1 DH AH0 AA1 D IY0 AH0 N S

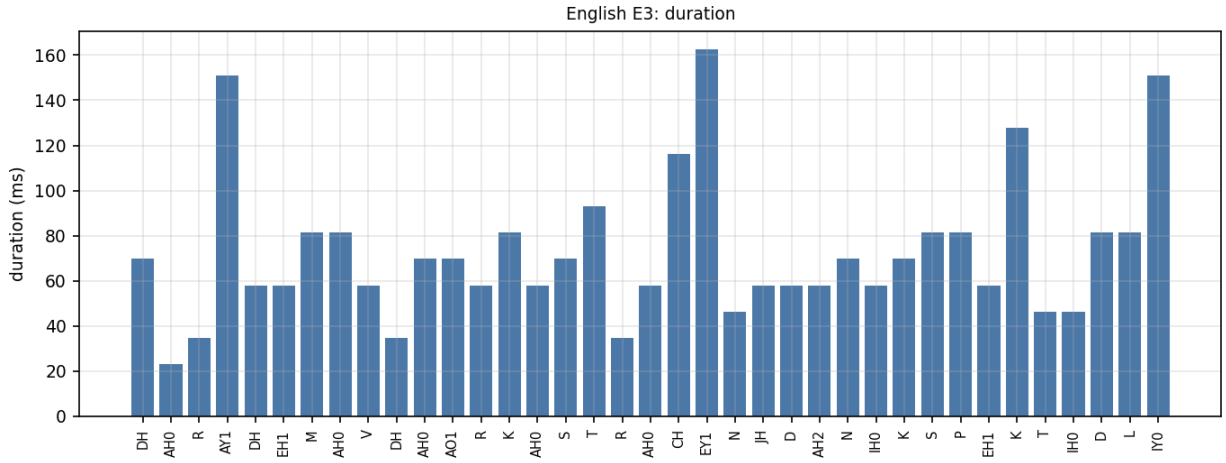
E2b

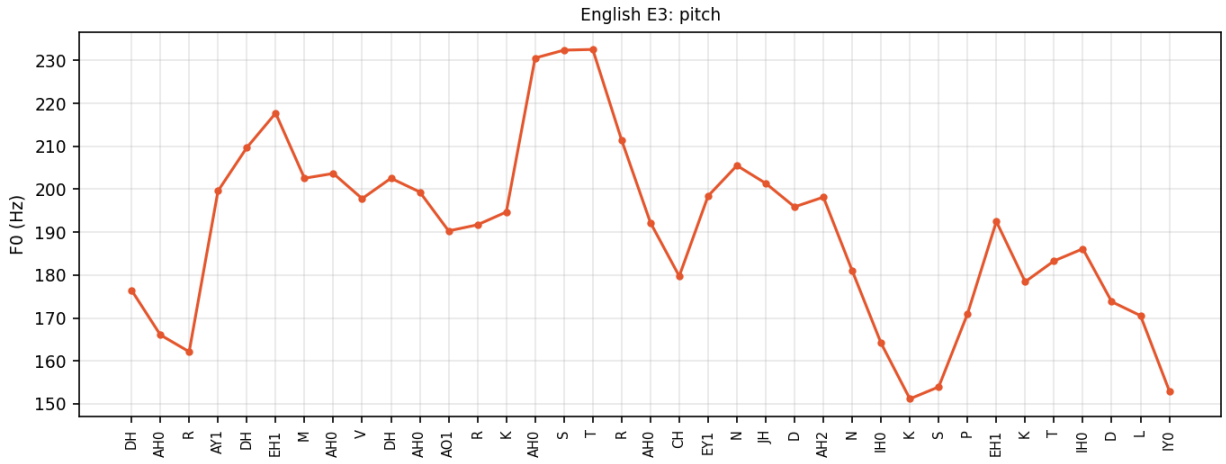

English E2b: duration
duration (ms)
0
20
40
60
80
100
120
140
DH AH0 R IY1 S ER0 CH ER0 Z W AH0 L K AA1 N D AH0 K T AE1 N IH0 K S P EH1 R AH0 M AH0 N T T AH0 IH0 V AE1 L Y UW0 EY2 T DH AH0 P R AH0 P OW1 Z D M EH1 TH AH0 D

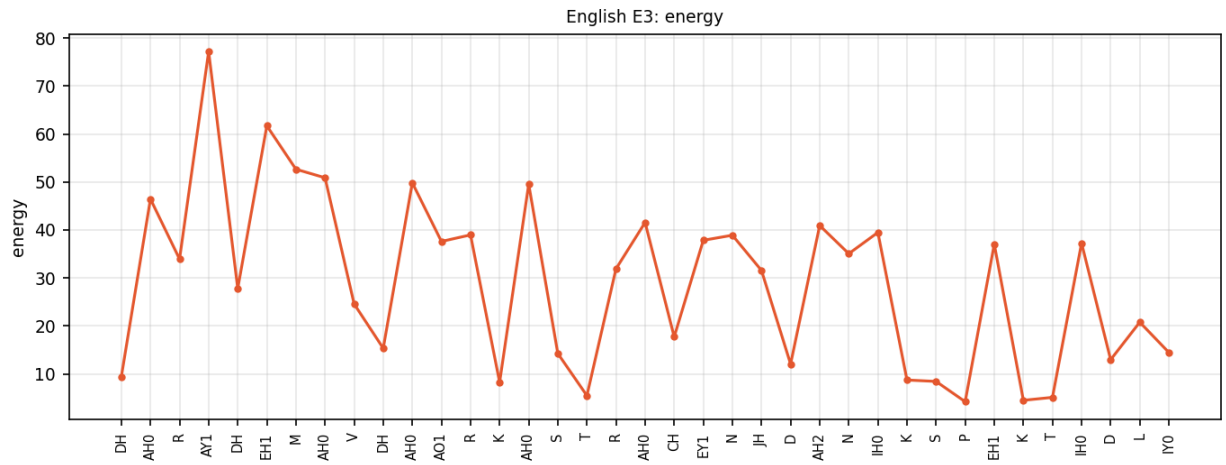

English E2b: pitch
F0 (Hz)
120
140
160
180
200
220
DH AH0 R IY1 S ER0 CH ER0 Z W AH0 L K AA1 N D AH0 K T AE1 N IH0 K S P EH1 R AH0 M AH0 N T T AH0 IH0 V AE1 L Y UW0 EY2 T DH AH0 P R AH0 P OW1 Z D M EH1 TH AH0 D


English E2b: energy
energy

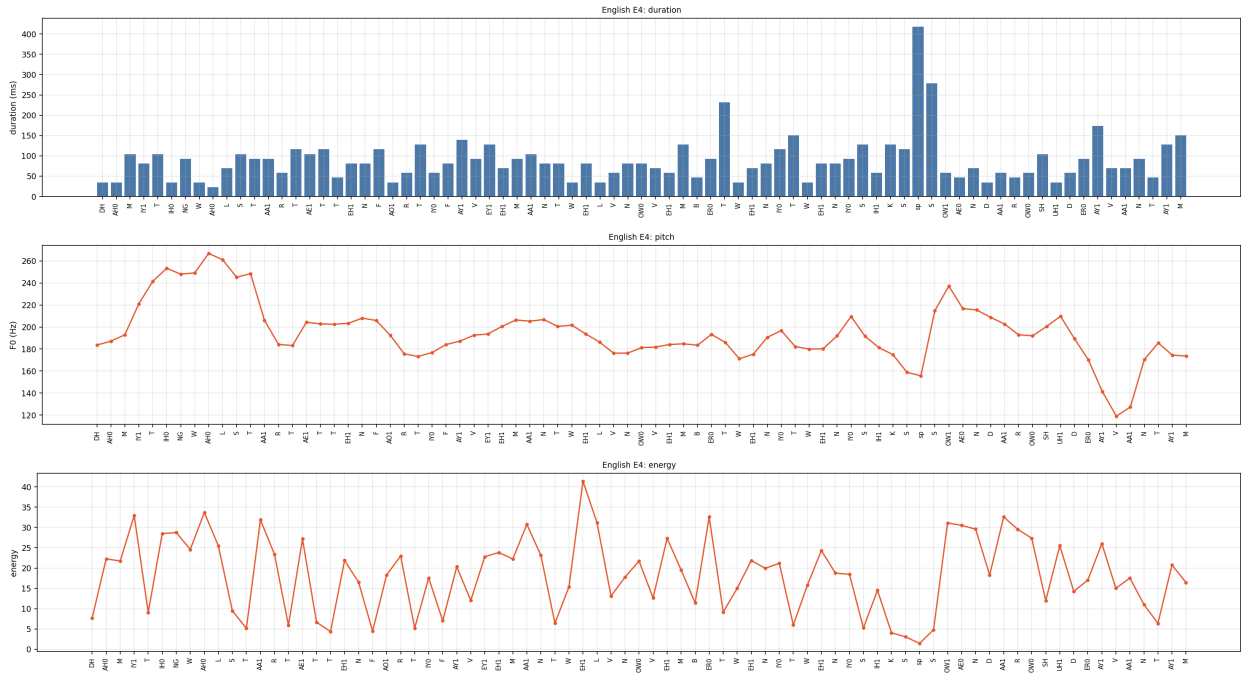
E3


English E3: duration
duration (ms)

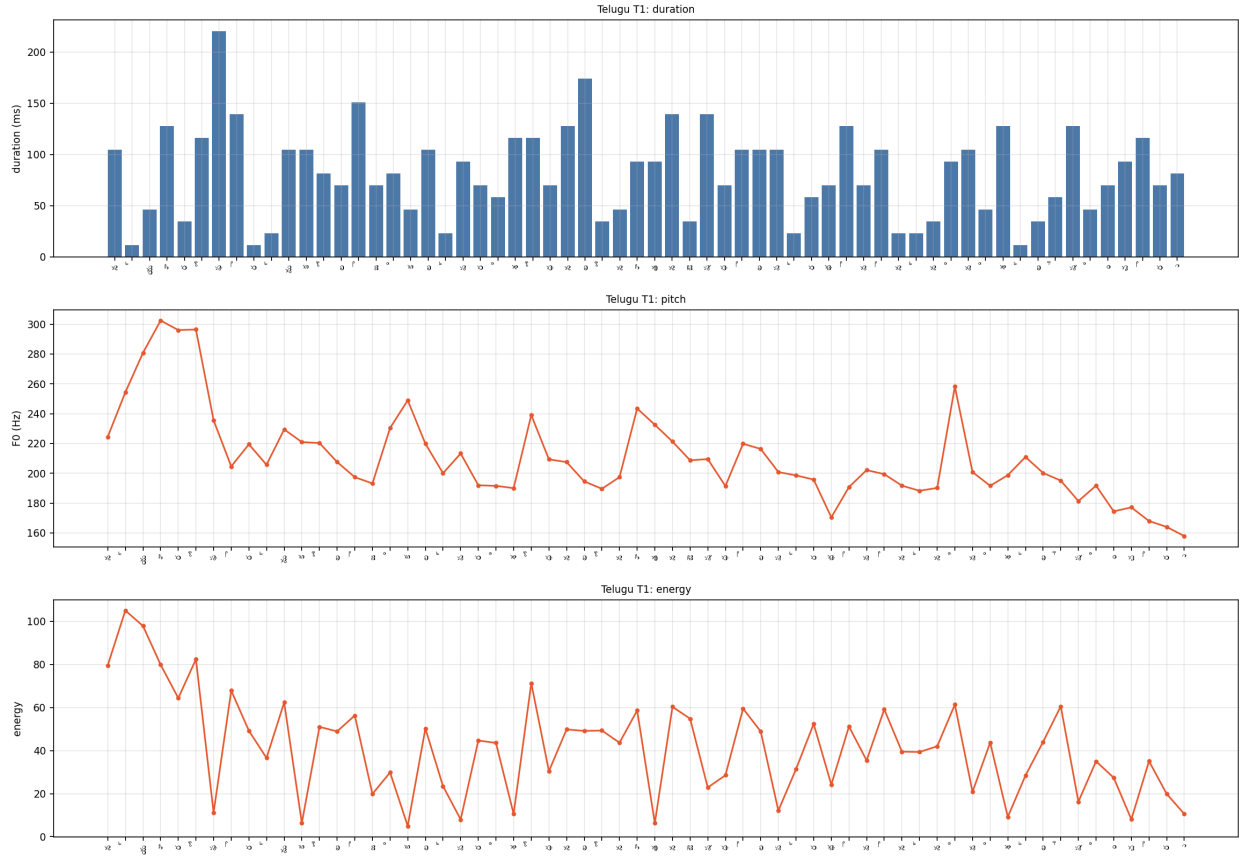

English E3: pitch
F0 (Hz)


English E3: energy
energy

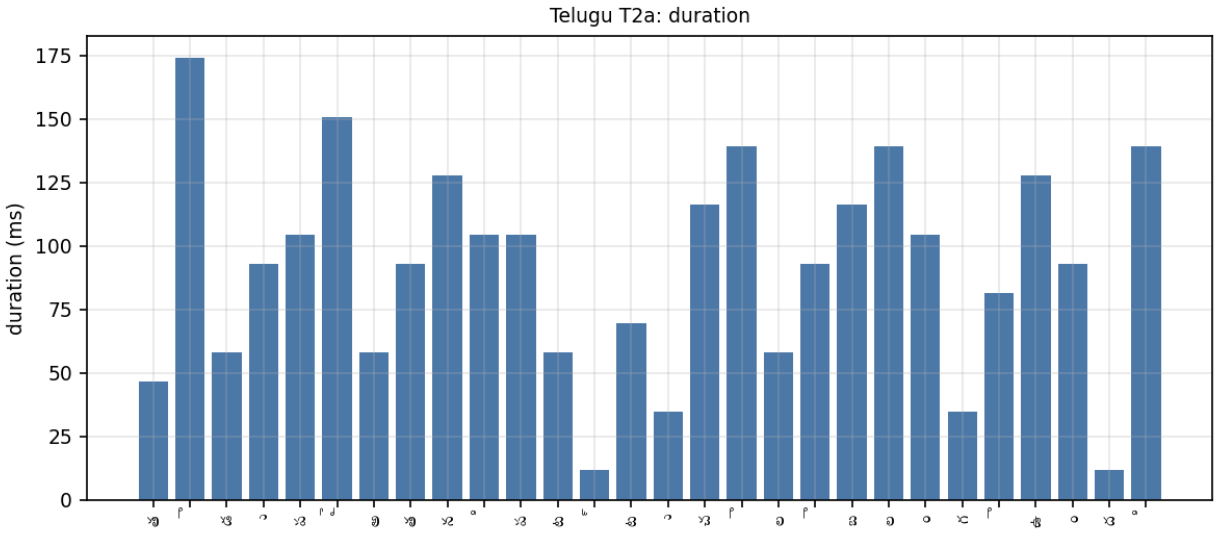
E4

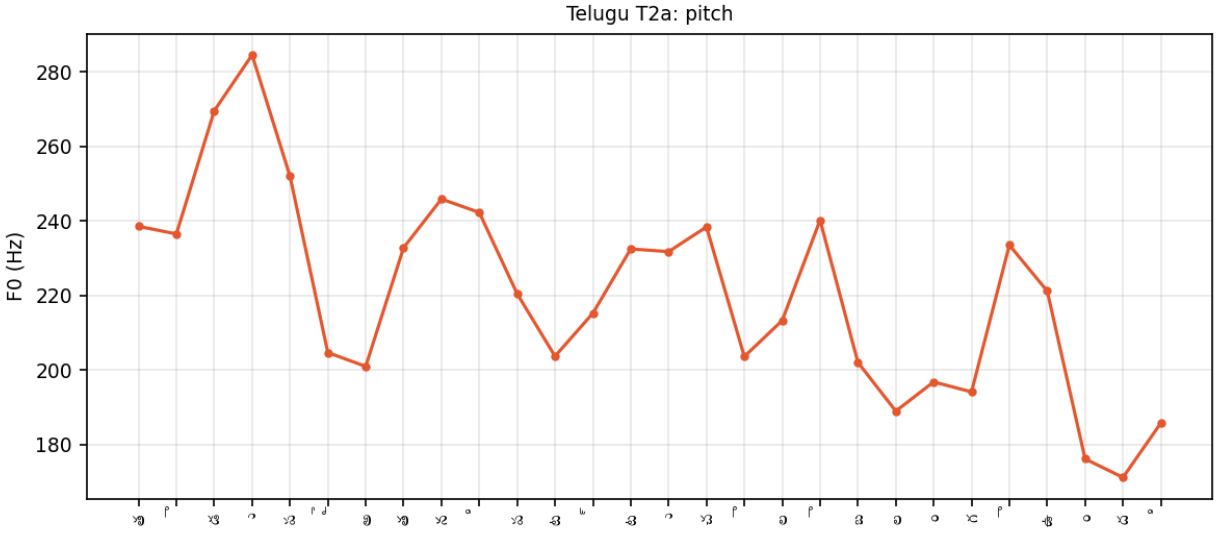

English E4: duration
duration (ms)
English E4: pitch
F0 (Hz)
English E4: energy
energy

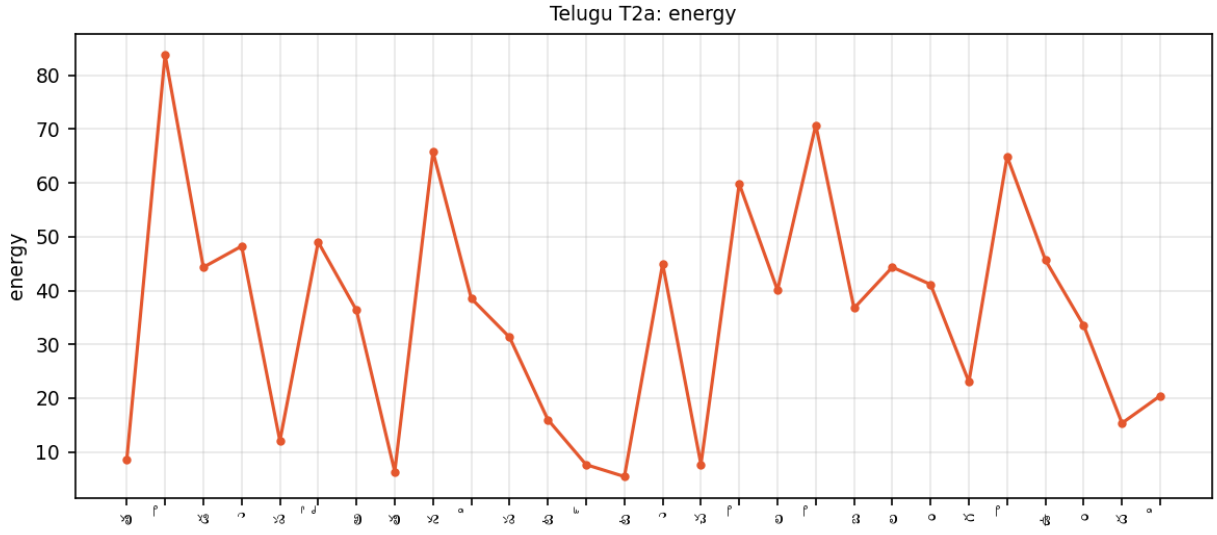
T1


Telugu T1: duration
duration (ms)
Telugu T1: pitch
F0 (Hz)
Telugu T1: energy
energy

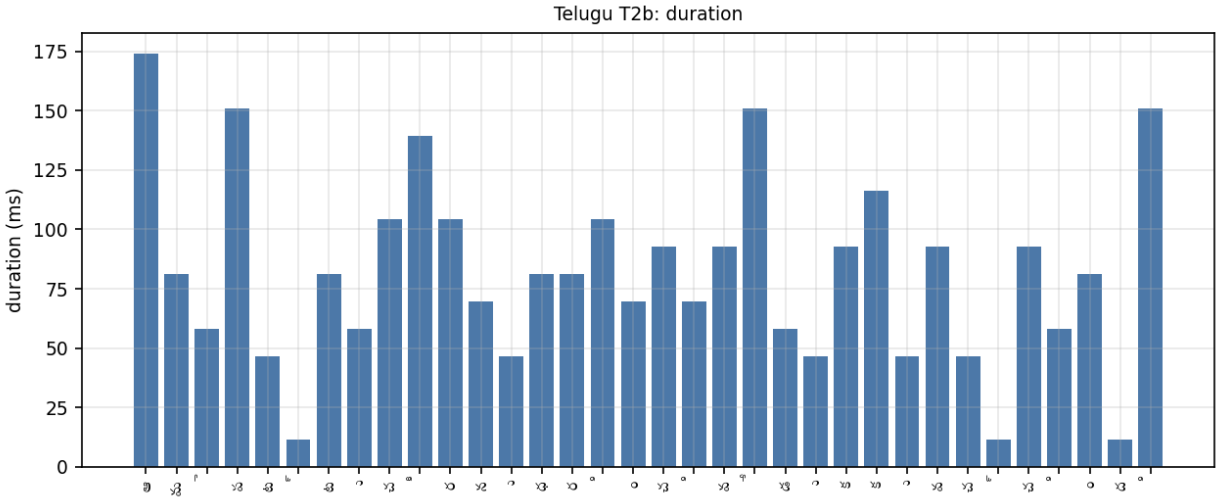
T2a

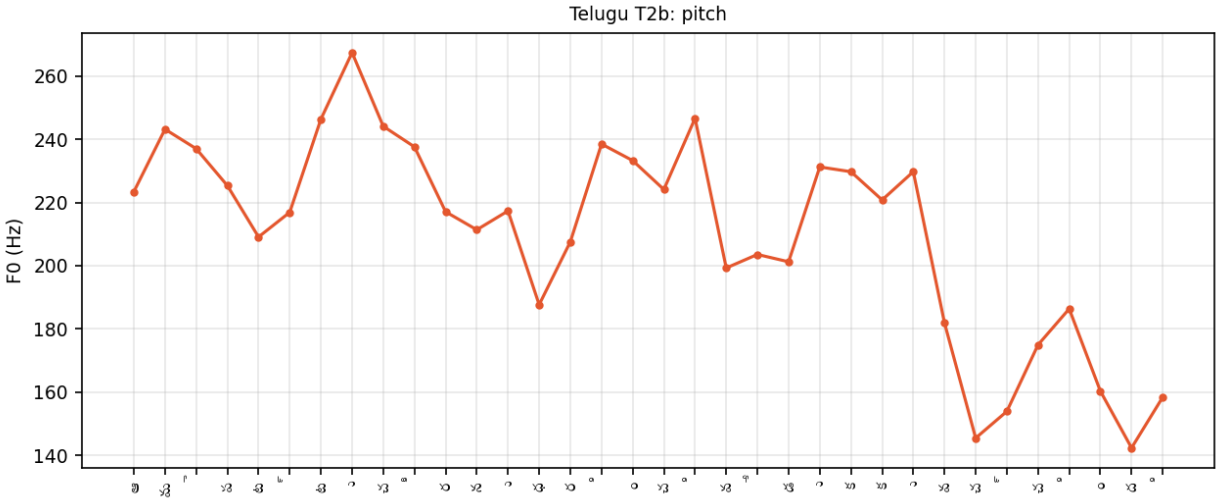

Telugu T2a: duration
duration (ms)
0
25
50
75
100
125
150
175

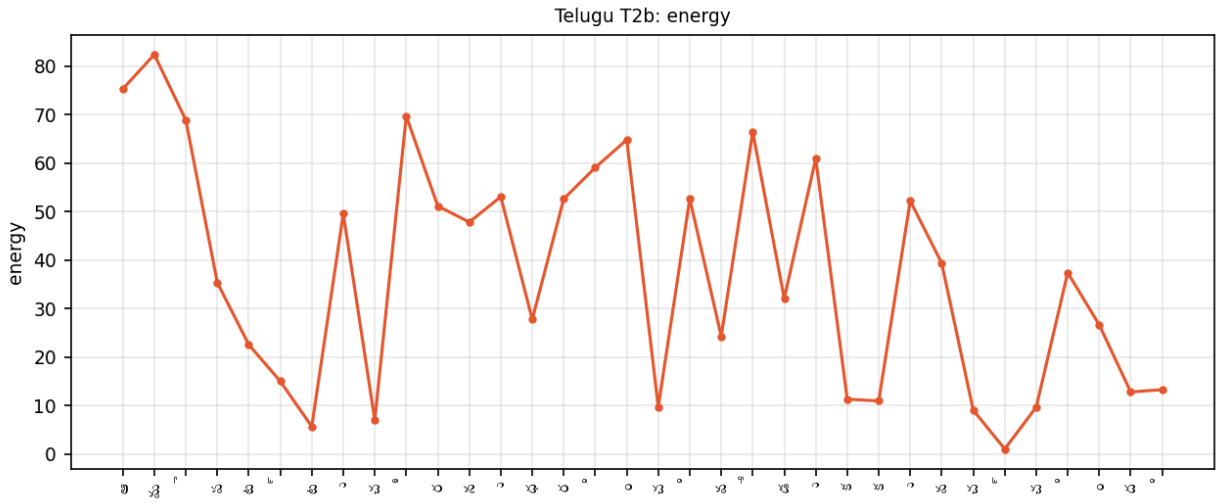

Telugu T2a: pitch
F0 (Hz)
180
200
220
240
260
280


Telugu T2a: energy
energy
10
20
30
40
50
60
70
80

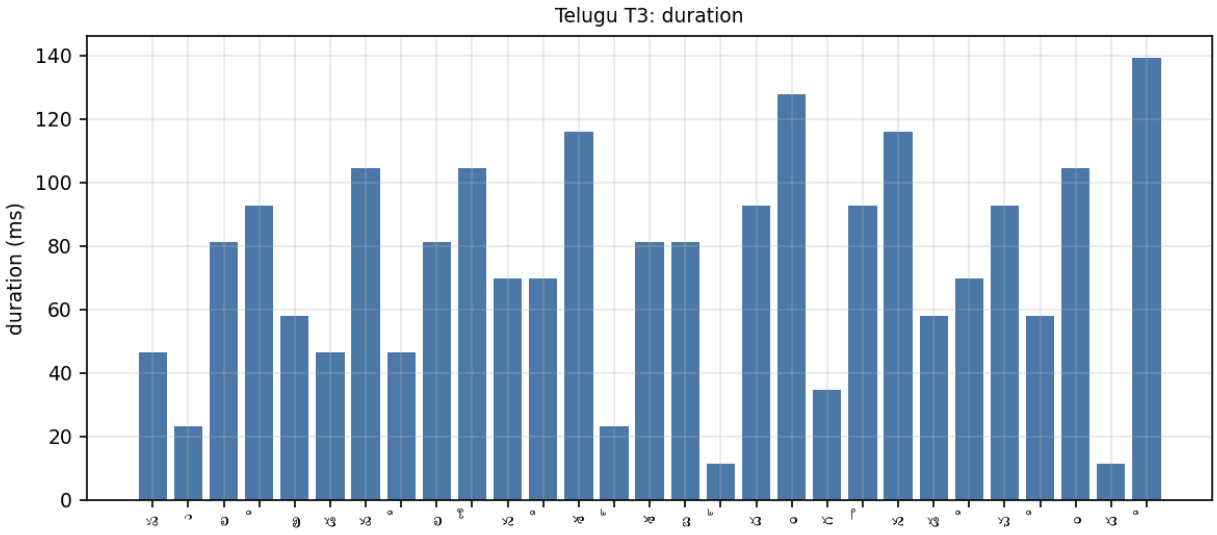
T2b

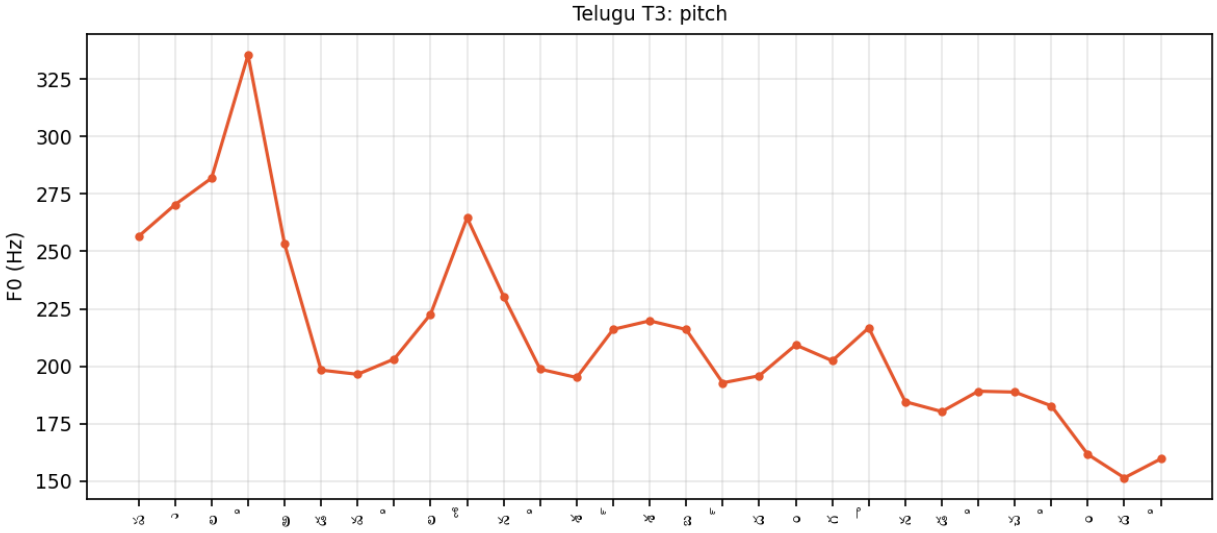

Telugu T2b: duration
duration (ms)
0
25
50
75
100
125
150
175

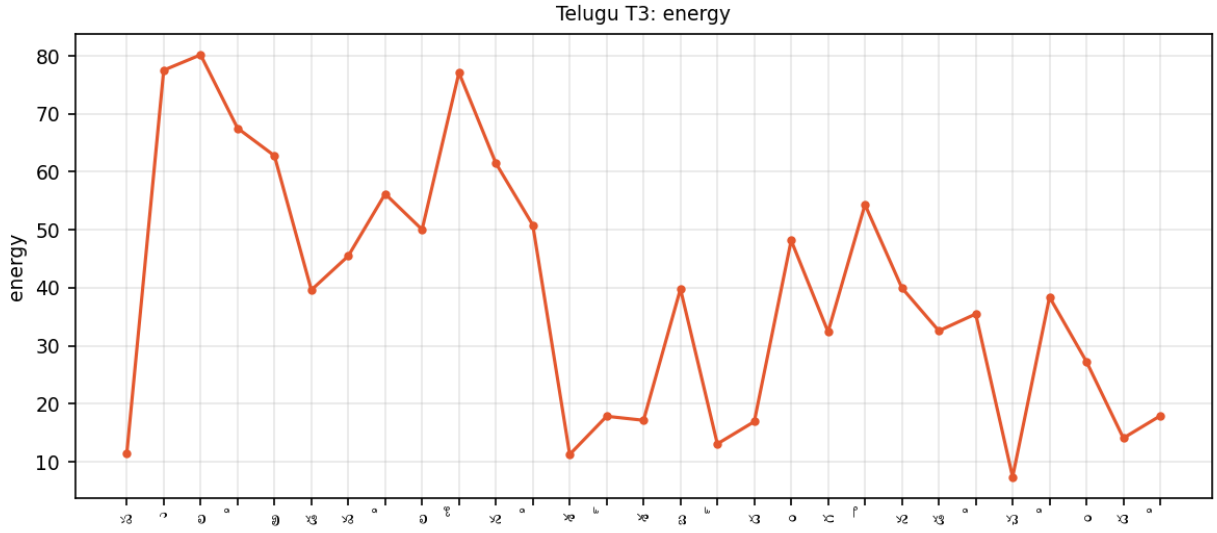

Telugu T2b: pitch
F0 (Hz)
140
160
180
200
220
240
260


Telugu T2b: energy
energy
0
10
20
30
40
50
60
70
80

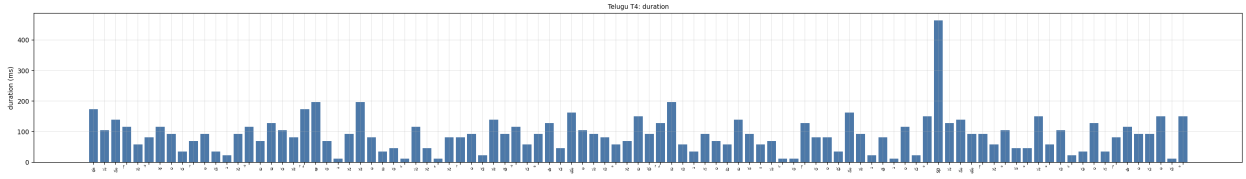
T3

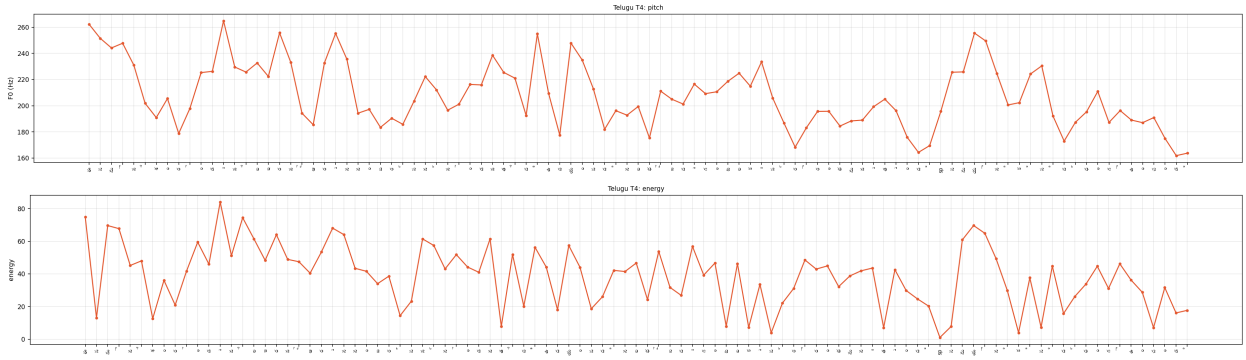

Telugu T3: duration
duration (ms)
0
20
40
60
80
100
120
140


Telugu T3: pitch
F0 (Hz)
150
175
200
225
250
275
300
325


Telugu T3: energy
energy
10
20
30
40
50
60
70
80

T4


Telugu T4: duration



6. English versus Telugu

Training loss is lower for Telugu (about 0.8 versus 1.05) and validation loss is much lower (about 2.2 versus 3.95). Telugu has cleaner, studio-quality, near-phonetic data, a grapheme dictionary that matches pronunciation closely, and only two speakers, so the variance targets are easier to predict. English comes from a noisy multi-speaker ASR corpus treated as one speaker, with irregular spelling handled through ARPAbet from a pronunciation lexicon and g2p, which gives noisier alignments and targets. Telugu pitch is higher on average (about 193 Hz versus 177 Hz) with a smaller spread; the Telugu model must also condition on a speaker embedding. The long words in E1 and T1 and the homographs in E2a/E2b and T2a/T2b test the front-ends: English relies on a lexicon plus g2p, while Telugu depends on a character-to-symbol mapping that cannot distinguish homographs by itself, so any difference comes from context learned by the encoder.

7. Limitations

English is treated as a single speaker although Svarah has many speakers, which probably explains the high validation loss. 35 English and 5 Telugu utterances were not aligned and were dropped. The Telugu aligner was trained for only 10 + 10 iterations to fit the compute limit. Models were trained once with 20,000 steps; no hyperparameter search or listening test with other people was done, and quality notes are from one listener.

8. Reproducibility

Repository: https://github.com/rashiii02/fastspeech2-english-telugu. Install with pip install -r requirements.txt (Python 3.13). Generate test audio with: cd English; python inference.py --ckpt checkpoints/English_20000_fp16.pth.tar, and cd Telugu; python inference.py --ckpt checkpoints/Telugu_20000_fp16.pth.tar. The full pipeline (data sampling with seed 42, alignment, TextGrid placement, preprocessing, train/val split and training) is in scripts/pipeline.py and per-language preprocess.py and train.py; each language folder has its config.yaml. Both checkpoints are in the repository under each language's checkpoints folder.